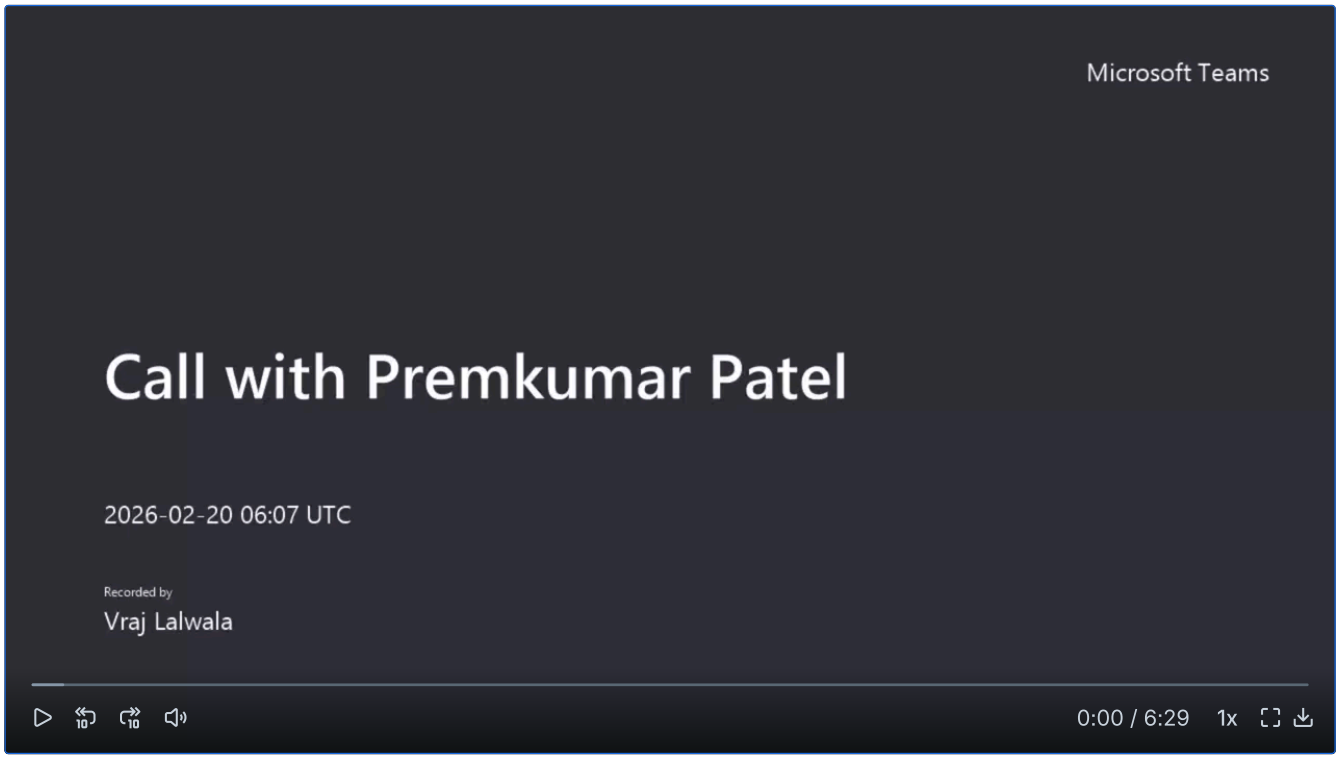
Comparison Report Task-4: Amazon S3 vs Amazon EC2 vs Amazon ECS



Frontend Deployment Comparison:-

Criteria	EC2 (Nginx)	ECS (Fargate + ALB)	S3 Static Hosting
Infrastructure Type	Virtual Machine (Server-based)	Container-based (Serverless compute)	Fully Serverless
Setup Complexity	Medium	High	Very Low
Operational Overhead	High (OS patches, security, scaling)	Medium (container management)	Very Low (managed by AWS)
Scalability	Manual (Auto Scaling required)	Automatic (Fargate scaling)	Automatic (built-in)
Cost	Medium (EC2 instance always running)	Higher (ALB + Fargate cost)	Lowest
Performance	Good	Very Good	Very Good (with CloudFront excellent)

High Availability	Must configure manually	Built-in with ALB & multiple tasks	Built-in
Security Management	Manual (security groups, updates)	Managed infrastructure	Minimal (policy-based)
Best For	Legacy apps, custom server logic	Microservices, scalable apps	Static frontend apps
DevOps Effort	Moderate	High	Minimal
CI/CD Friendly	Yes	Yes (best for containers)	Yes (very simple deployment)
Path-Based Routing Support	Via Nginx config	Via ALB rules	Not required
Server Maintenance	Required	Not required (Fargate)	Not required
Docker Required	No	Yes	No
Learning Curve	Low–Medium	High	Very Low

Summary of Advantages & Disadvantages

EC2 (Nginx)

Advantages

- Full control over server
- Flexible configuration
- Easy to debug
- Good for traditional hosting

Disadvantages

- Requires server maintenance
- Manual scaling
- Higher operational responsibility
- Security patching required

ECS (Fargate + ALB)

Advantages

- Highly scalable
- Containerized architecture
- Production-ready
- Good for microservices
- No server management (Fargate)

Disadvantages

- More complex setup
- Higher cost (ALB + Fargate)
- Requires Docker knowledge
- Path routing complexity

S3 Static Hosting

Advantages

- Cheapest option
- Fully serverless
- Automatic scaling
- Very simple deployment
- No server or container management

Disadvantages

- Only supports static content
- No backend logic
- Limited server-side customization

Final Recommendation:-

For static frontend applications like:

- Air-Quality
- Bookshop
- Notesapp

The recommended standard service is:

> ### Amazon S3 Static Website Hosting

Because:

- Minimal operational overhead
- Lowest cost
- Auto scaling
- Simple deployment
- No infrastructure management required

For enterprise-scale containerized environments → ECS is recommended.

For traditional hosting environments → EC2 can be used.